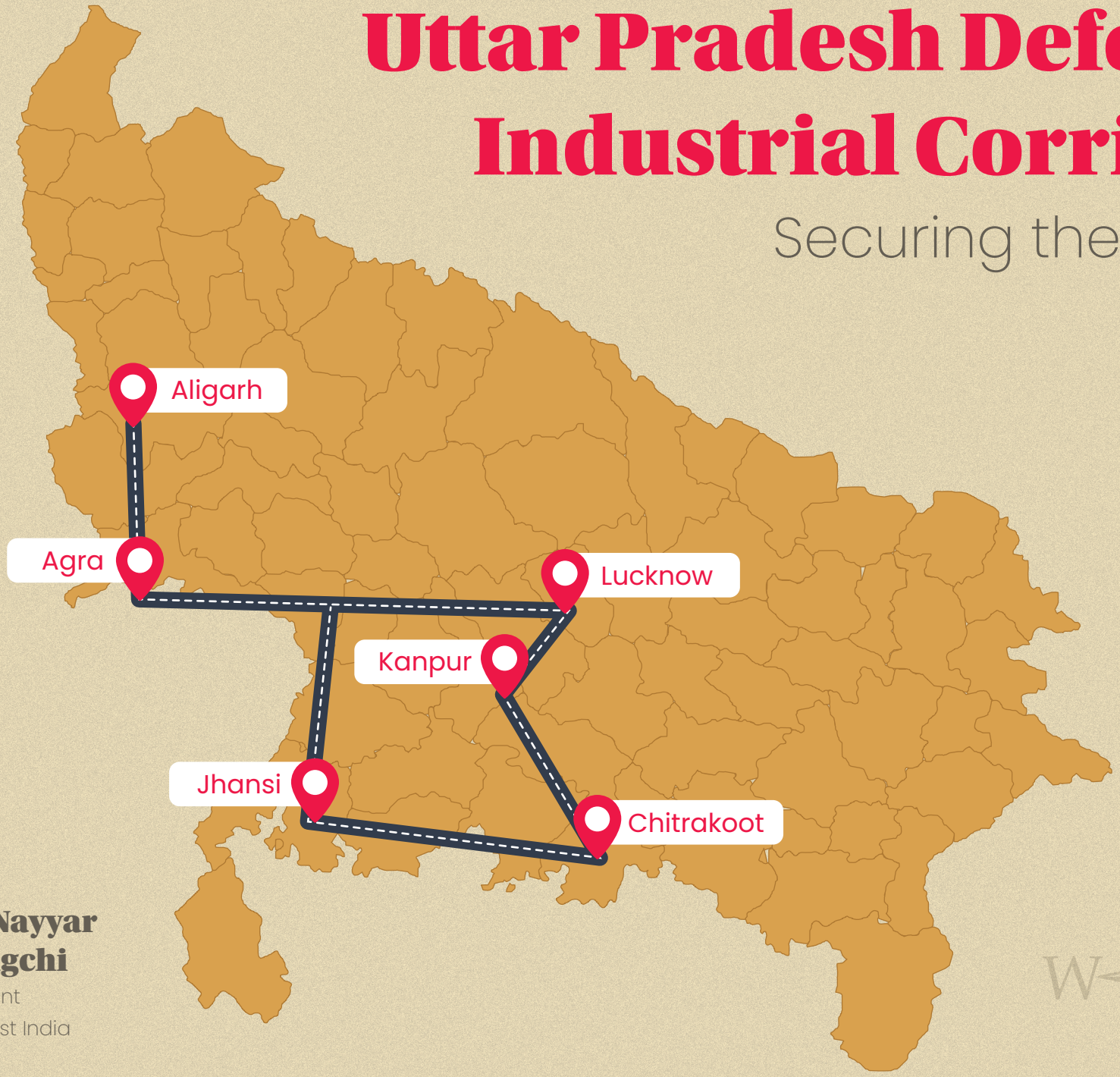
Uttar Pradesh Defence Industrial Corridor:

Securing the Nation



By Mishika Nayyar and Vivek Sigchi

Strategic Investment Research Unit, Invest India

Uttar Pradesh Defence Industrial Corridor

Securing the Nation

उत्तर प्रदेश सरकार

upeida
UP Expressways Industrial Development Authority

INVEST INDIA
NATIONAL INVESTMENT PROMOTION & FACILITATION AGENCY

Yogi Adityanath

Hon'ble Chief Minister, Uttar Pradesh

Message

It's with pleasure that I announce the setting up of the Defence Industrial Corridor in the state of Uttar Pradesh pursuant to the announcement made by the Hon'ble Prime Minster Shri Narendra Modi Ji in the investors summit held on 21 February 2018.

Uttar Pradesh has significantly grown its infrastructure over the past few years. There are ideal locations in the state that have been carved out specifically for the defence industry. Moreover, the government organized the Defence Expo 2020 in Lucknow to showcase the preparedness of our country to attract investments in the sector. 'The Uttar Pradesh Defence and Aerospace unit and Employment Promotion Policy' is already in place which offers many incentives for potential investors in addition to infrastructural facilities and logistics support. We welcome all global and domestic investors to come and invest in the state.

Aiming to showcase the details of the Defence Industrial Corridor, I would like to congratulate the Strategic Investment Research Unit at Invest India for coming up with a comprehensive report on Uttar Pradesh Defence Corridor: Securing the Nation. Hope this serves as a good repository of information for all the involved stakeholders. The state is committed to providing the best possible support to everyone involved.

Yogi Adityanath
Chief Minister, Government of Uttar Pradesh

Satish Mahana

Uttar Pradesh Industries Minister

Satish Mahana
MINISTER
Industrial Development Department

Office :
Room No. : 74-74 A Main Building
U.P. Secretariat
Phone: 0522-2238388
Lucknow Date.....

Message

India is among the top 5 military spenders and one of the emerging defense manufacturing hubs in the world. To support the growth of the defense sector and enhance manufacturing capacity in the sector, we are setting up a Defense Industrial Corridor in the state. This project shall put Uttar Pradesh on the world map as one of the key defense manufacturing hubs in the world.

Uttar Pradesh can meet all the requirements of being a perfect investment destination in terms of infrastructure, well-developed logistics and a conducive business environment. The state of Uttar Pradesh has also ensured good connectivity, round-the-clock power supply and robust law and order situation. Thus, making it a sound destination for carrying out a venture of this scale.

The report further highlights the strengths of the state in achieving this vision. I would like to congratulate the Strategic Investment Research Unit at Invest India for coming up with a comprehensive report on Uttar Pradesh Defence Corridor: Securing the Nation.

(Satish Mahana)

(Satish Mahana)
Minister
Industrial Development Dept.
Uttar Pradesh

Awanish Kumar Awasthi

CEO, UPEIDA

उत्तर प्रदेश एक्सप्रेसवेज औद्योगिक विकास प्राधिकरण
UP Expressway Industrial Development Authority
सी-13, पर्यटन भवन, द्वितीय तल, विपिन खण्ड, गोमती नगर, लखनऊ-226010
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Invest India
New Delhi

Message

UPEIDA has been working relentlessly towards increasing the state's infrastructure and fast tracking its progress. Building a Defence Industrial Corridor is one such initiative that can bring great benefits to the economy, as was announced by Hon'ble Prime Minister Narendra Modi in February 2018.Of one of the two defence industrial corridors announced, the one proposed in the Bundelkhand region of the Uttar Pradesh will not onlyhelp the MSME sector but also attract huge investments along with generating employment for 2.5 lakh people, especially for the youth of Bundelkhand.

Uttar Pradesh Defence Industrial Corridor (UPDIC) is an aspirational project that intends to provide an impetus to the defence production industry in the countryand reduce foreign dependency of Indian Aerospace & Defence Sector. The project shall also help in attracting huge investments in the defence sector in Uttar Pradesh, which hold immense potential in terms of the opportunities available and existing favourable set-up. Moreover, our industry friendly 'U.P. Defence and Aerospace Unit Employment Promotion Policy – 2018' is in place to help the industry grow and add value to the economy of the state and the country. The state of Uttar Pradesh is committed to provide a conducive environment for the development of defence industries and pledges all possible support to investors.

I hope the readers find this report on informative which highlights this aspirational project. Congratulations to the Strategic Investment Research Unit at Invest India for coming up with this report on **Uttar Pradesh Defence Corridor: Securing the Nation.**

Awanish Kumar Awasthi
Chief Executive Office
UPEIDA

Deepak Bagla

Managing Director and CEO, Invest India

Message

With the growing demand in the defence sector the government has identified it as one of the core areas to boost 'Atma Nirbhar Bharat or Self-Reliant India.

The Government has also spelt out its vision of achieving a turnover of USD 25 Bn including export of USD 5 Bn in Aerospace and Defence goods and services by 2025. To support defence modernisation, the governments at the centre and states, especially Uttar Pradesh, has called for increased participation of defence manufacturing companies from the private sector to cater to the growing demand. The upcoming defence industrial corridor in Uttar Pradesh and opening up of the Indian defence industry also paves the way for foreign original equipment manufacturers to enter strategic partnerships with defence equipment manufacturers in India.

The upcoming defence industrial corridor in Uttar Pradesh will continue to encourage investments in Uttar Pradesh which have been flowing in huge numbers especially with recent announcements made by DRDO Brahmos Aerospace, Bharat Dynamics and many more.

Showcasing the upcoming defence industrial corridor in Uttar Pradesh will be a key driver in securing the nation. This report by Strategic Investment Research Unit by Invest India on Uttar Pradesh Defence Industrial Corridor will also serve as a repository of information for all potential investors.

Introduction

India is among the top 5 military spenders and one of the emerging defence manufacturing hubs in the world. To support the growth of the defence sector and enhance manufacturing capacity in the sector, two Defence Industrial Corridors are being set up in India, one in Uttar Pradesh and another in Tamil Nadu. A defence corridor refers to a route or a path along which domestic production of defence equipment by public sector, private sector and MSMEs are lined up to enhance the operational capacity of the defence forces. **Promoting Make in India, the Defence Industrial Corridors will catalyse indigenous production of defence and aerospace-related items.** This will reduce our imports and promote the export of these items to other countries. The combined efforts of the Government and private players will help achieve India's goal of self-reliance in defence, generate direct and indirect employment opportunities and spur the growth of private domestic manufacturers, Micro Small and Medium Enterprises (MSMEs) and start-ups.

Defence manufacturing in India

India accounts for 3.7% of the global military spending, making it the third highest military spender in the world. The defence expenditure constituted 2.9% of India's total GDP with approximately USD 72.9 billion spent on defence in 2020. Under the Field Artillery Rationalization Plan - 1,580 towed guns, 100 tracked guns, 814 Mounted Gun System will be required in India.

With the growing demand in the defence sector the government has identified it as one of the core areas to boost 'Atma Nirbhar Bharat or Self-Reliant India'. The Government has also spelt out its vision of achieving a turnover of USD 25Bn including export of USD 5 Bn in Aerospace and defence goods and services by 2025.
To support defence modernisation, in the budget 2021-22 the defence capital outlay has been

increased by 18.75% from the preceding year. This is the highest ever increment in the last 15 years.

To cater to the growing demand, the government has also called for an increased participation of the private sector in defence manufacturing. The opening up of the industry also paves the way for foreign original equipment manufacturers to enter strategic partnerships with Indian companies.

The government aims to ensure transparency, predictability, and ease of doing business by creating a robust eco-system and supportive government policies. Towards this end the government has taken steps to bring about de-licensing, de-regulation, export promotion and foreign investment liberalization. To meet export requirements, a draft Production and Export Promotion Policy (DPEPP) 2020 has also been formulated.

The Government of India has enhanced FDI in Defence Sector up to 74% through the Automatic Route for companies seeking new defence industrial license and up to 100% by Government Route.

Modernization of armed forces
$ 130 bn opportunity in coming 7-8 years

Indigenously designed, developed and manufactured (IDDM)
New category of capital procurement

R&D
$ 15.4 mn allocated to setup 'Technology Development Fund'

Positive Indigenisation list for local procurement
This will boost indigenisation with active participation of public and private sector for achieving self-reliance and promoting defence exports. 1st List (101 items) and 2nd List 108 items)

Defence Sector : An Overview

DEFENCE BUDGET
- 2 % of GDP spent defence
- US$67.8 B

FDI POLICY
74% Automatic Route;
100% Govt. Route

BORDERS
- 7,517 kms of coastline
- 15,000 kms of land borders

EXPORTS
- $1.3 Bn worth of exports (2019-20)
- Exports to 68 countries (2019-20)

PRODUCTION
- 12 Billion USD Defence & Aerospace economy
- Private Sector licenses doubled to 452 in 5 years

OFFSETS
30% offset obligation is required for procurement

ARMED FORCES
- 2nd largest armed force
- 1.3 Mn active personnel

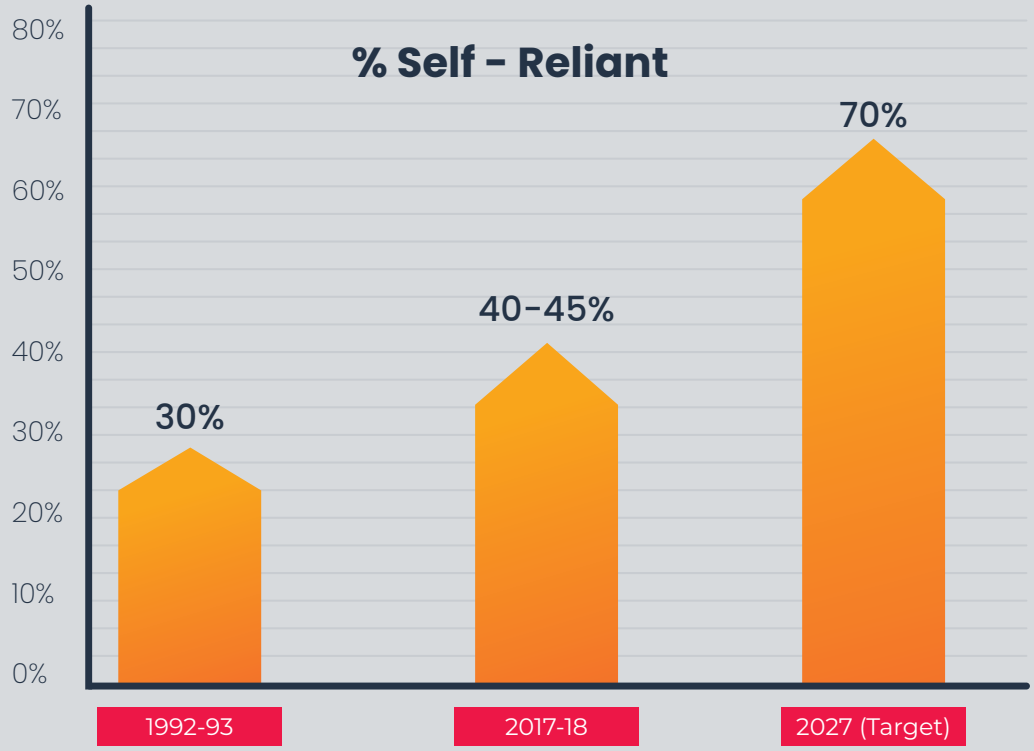
MILITARY MODERNIZATION
- $130 Bn requirement
- US$ 19 Bn for capital expenditure (2020-21)

India's Self Reliance Index

- Increasing levels of **self-reliance** in defence production
- **All naval ships & submarines** are built in India
- **75%** of the total acquisition orders of the Indian Army are with **Indian firms**
- Further increase in self-reliance targeted
- **~70%** self-reliance target by 2027

Under Atmanirbhar Bharat

- 101 items to be manufactured domestically for defence sector, instead of being sourced via imports.
- The government created a separate budget for domestic capital procurement in the current financial year.



Make in India (launched Sept'14)

Overview of DPSUs

9 Defence Public Sector Undertakings

7 Newly Formed DPSUs from Corporatisation of 41 Ordnance Factories

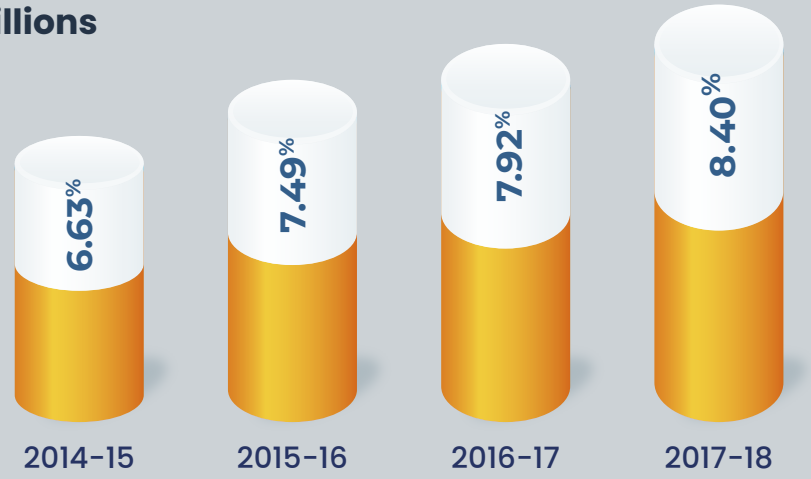
~$9B In value of production in (2018-19)

Product Range

- 155 M Artillery, surface-to-air weapon systems, strategic weapons, launchers, torpedos, decoys and missiles
- Missile destroyers, stealth frigates and Scorpene submarines, warships, landing craft utility, warfare corvettes
- Avionics, weapons systems, electronic warfare and naval systems, radars, etc.
- Multi-role fighters, Hawk - advanced jet trainer, light combat aircraft (LCA), light transport aircraft, helicopters, etc.

6% growth per annum since 2015

$ Billions



Note: All figures converted from INR using Aug 13 exchange rate

1. Roads & Highways, Invest India

Overview of DPSUs

490 License issued,

42 Joint ventures with foreign companies

$2.2B In value of production in (2018-19)

Product Range

- Electronic Warfare systems, Radars and avionics systems, communication systems
- Optical goods, navigational instruments, night vision devices
- Logistic vehicles
- Rocket launchers, sensors, simulators, Ammunition hardware, Mechanical Fuses
- Protection equipment, safety and ballistic products

Private Sector Partners

adani Defence and Aerospace

TATA ADVANCED SYSTEMS

LARSEN & TOUBRO

mahindra

RELIANCE Reliance Defence and Engineering Limited

Godrej AEROSPACE

BrahMos

KALYANI

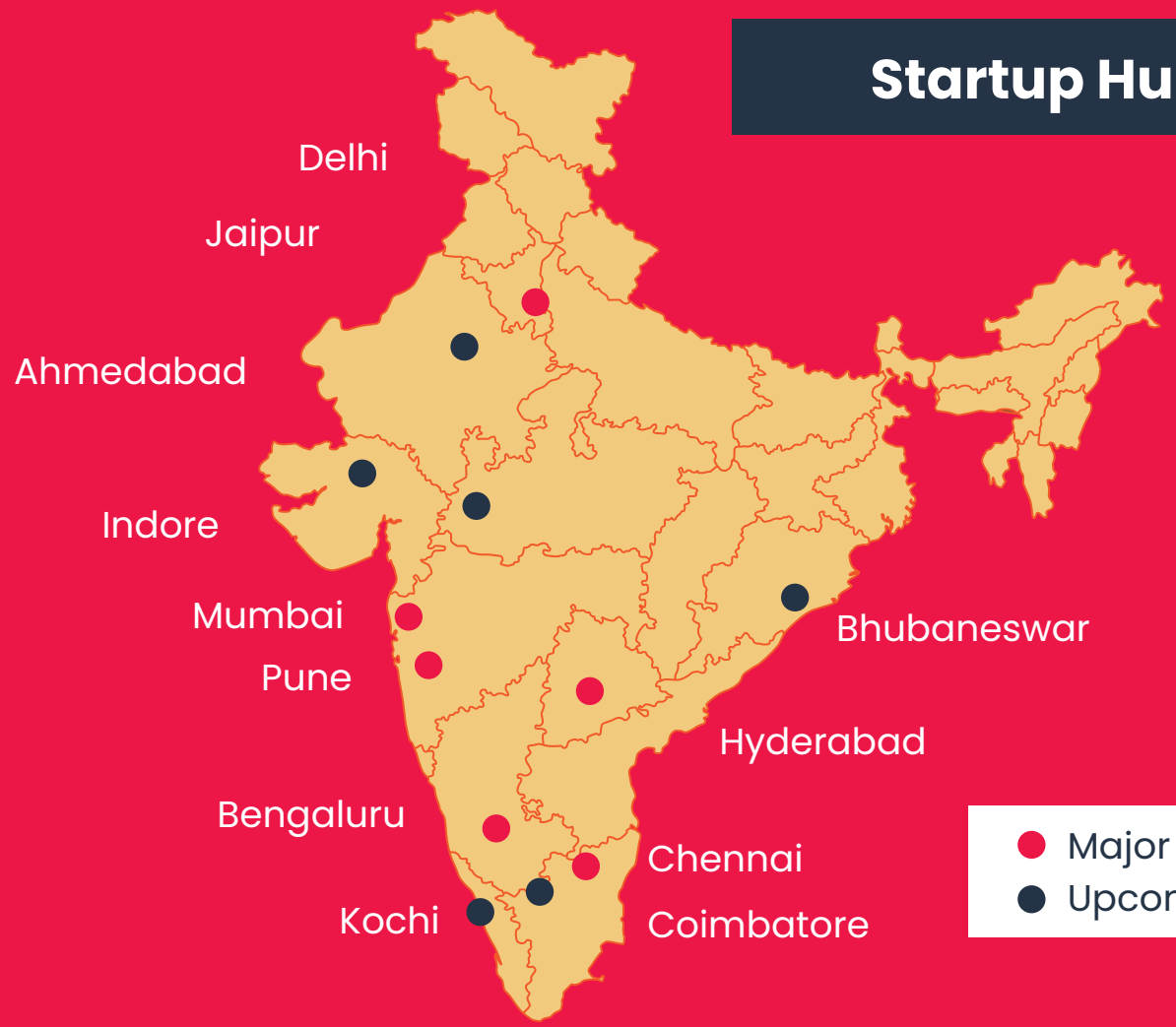
3rd Largest Startup Ecosystem

20,000+
Startups in India

$50 Bn
Valuation

200+
Incubators/ Accelerators

$172 Bn
Funding since Jan'16



28 Yrs
Avg. age of founders

40%
Incubator Annual Growth

9%
Women
(100% growth in last
12 months)

700+
Startups arising from
unicorns

Defence Startups

Contracts extended to **33 innovators**

Grant of USD **5.4 M** sanctioned

iDEX Innovations for Defence Excellence

- Fostering innovation and development in the A&D Sector and increasing self reliance & indigenization
- Providing diverse stakeholders with funding/other support to carry out R&D development
- Additional Rs 500 Crore approved for iDEX In June 2021

Advantage UP

Strategically located on the Golden Quadrilateral, the state is connected to all major national and international airports in the country. The state also has the largest railway network in the country spanning 8,949 km. Western Dedicated Freight Corridor – WDFC, being developed from Dadri in Ghaziabad to Jawaharlal Nehru Port in Mumbai will lead to a huge reduction in transportation time giving major boost to the economic activity in the state. Likewise, Eastern Dedicated Freight Corridor – EDFC will help in increasing productivity and enhancing output. The state government is also encouraging the development of integrated manufacturing clusters, logistics and integrated industrial townships in the cities adjacent to these corridors, such as Greater Noida, Allahabad, Kanpur, etc. to derive maximum benefit out of these freight corridors.

The existing logistics infrastructure facilities in Uttar Pradesh, the joint domestic rail and EXIM (import-export) terminal at Moradabad, the rail-connected private freight terminal at Kanpur and the non-commercial container depot-ICD, including ICD at Kanpur and ICD located at Dadri terminal will also further the state's growth objectives. Additionally, three multi-nodal logistics / transport hubs are also proposed in the state in Noida, Bodaki and Varanasi. Apart from the special investment centres like Kanpur, Noida, Varanasi and Ghaziabad, new investment areas like Dadri-Noida-Ghaziabad special area, Meerut-Muzaffarnagar special area, Deendayal Upadhyay Nagar (Mughalsarai)–Varanasi-Mirzapur special area are also being developed. Moreover, for the purposes of connectivity, several expressways, four lane and six lane national highways, national waterways – 1 (NW-1), connecting international airports, Allahabad, Varanasi and Haldia Ports, are available. This will help industries & manufacturers of the state to transport their products efficiently using excellent facilities.

20 sectoral policies complementing UP Defence & Aerospace and Employment Promotion Policy (First amendment), 2019
State ranked 2nd in Ease of Doing Business
Time-bound clearances guaranteed through UP Janhit Guarantee Act

2nd largest economy of India
Highest number of MSMEs in the country
Highest consumer base with 16.5% of the country's population
Largest exporter in the country with 18% year-on-year growth

Uttar Pradesh is **fourth** largest state and **second** largest **economy in India**.

State consists of **16.5%** Population of the country.

State amongst **top Five Manufacturing** states of India

Uttar Pradesh **ranks first** in terms of **MSMEs** in India.

Defence Manufacturing in Uttar Pradesh – a Snapshot

Home to prominent PSUs:

- HAL manufacturing units,
- Ordnance Factories,
- Bharat Electronics Ltd., etc.

R&D and Innovation Base:

- Prominent academic institutes like IIT-Kanpur,IIT-BHU, etc.
- Key R&D institutes viz. DRDO, ASERDC-HAL*, etc.
- Prominent Forensics Science Laboratories

- Strong skilled manpower base
- 56% of UP's 220 million population in the working age group
- 53 Universities, 4345 Colleges, 168 Polytechnics

- Strong ancillary base
- Home to country's largest MSME base (around 9 million) with huge numbers of repairing/ servicing, engineering & spare parts manufacturing industry

Uttar Pradesh Defence Industrial Corridor

In the UP Investors Summit held in the month of February 2018, the Government of India announced the development of Defence Corridor in the Bundelkhand region of Uttar Pradesh with an investment of Rs. 20,000 crores. Uttar Pradesh Defence Industrial Corridor (UPDIC) is an aspirational project that intends to reduce foreign dependency of Indian Aerospace & Defence Sector. It is estimated that the construction of this corridor will generate more than one lakh employment opportunities. The Uttar Pradesh Expressways Industrial Development Authority (UPEIDA) was made the nodal agency responsible for executing this project in conjunction with various other state agencies. The project is being planned across 6 nodes namely – Lucknow, Kanpur, Jhansi, Agra Aligarh, Chitrakoot, which spreads across Central, East, West region of Uttar Pradesh and along the Golden Quadrilateral connecting Delhi – Kolkata supported by networks of expressways. There is a strong support base in these districts to meet the requirements of defence manufacturing and to ensure supply of raw material, labor, etc. The corridor is also in a special advantage due to its proximity to the DMIC and the ACIC. The Corridor aims to bring up the state as one of the largest & advanced defence manufacturing hubs and put it on the world map.

Uttar Pradesh Defence Industrial Corridor

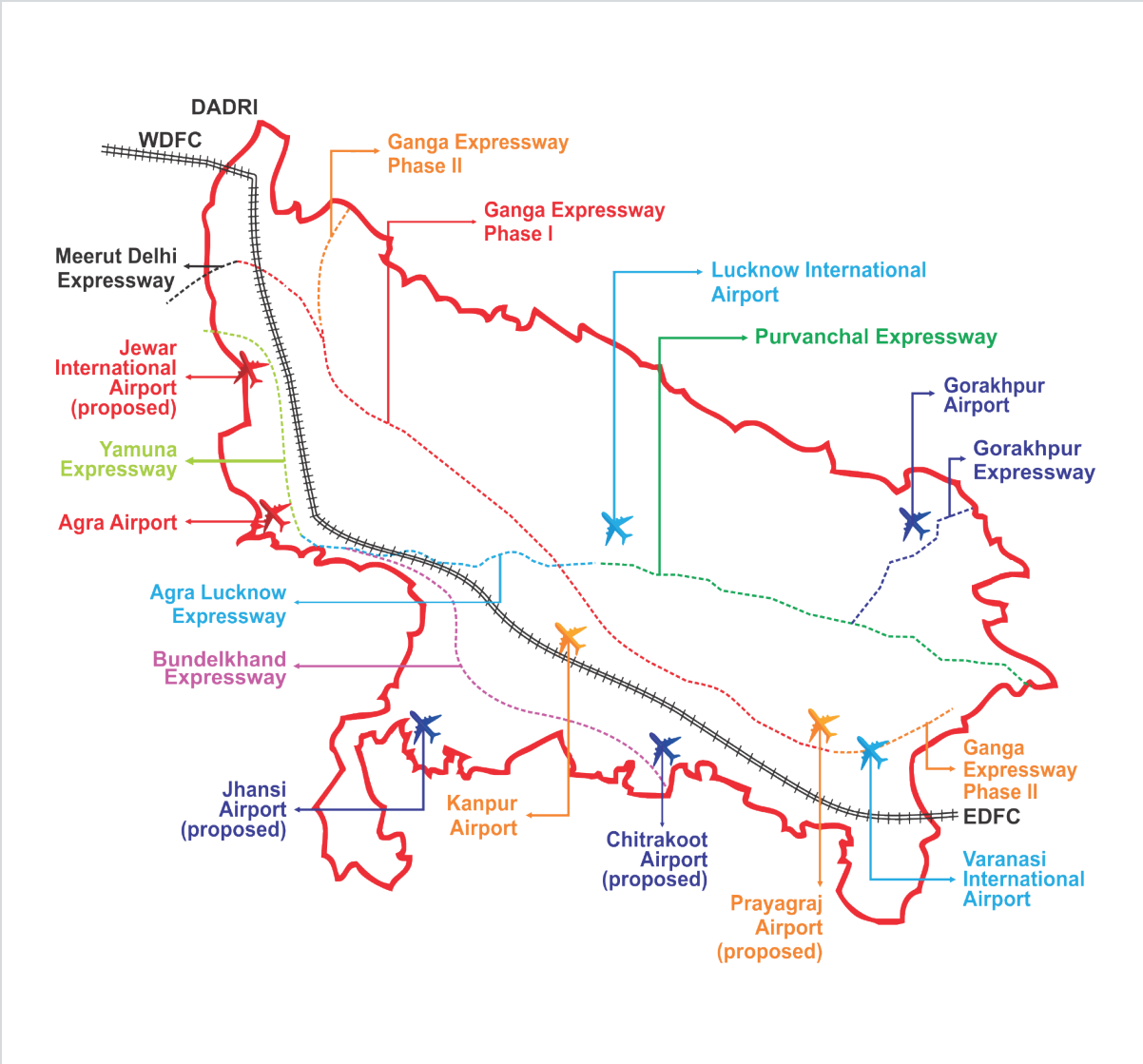
- Announcement of setting up of two Defence Industrial Corridors on 1st February 2018 by Hon'ble PM Shri Narendra Modi Ji

  Defence Industrial Corridors established in:
  - Uttar Pradesh
  - Tamil Nadu

- UP Defence Industrial Corridor planned across six nodes – Agra, Aligarh, Lucknow, Kanpur, Chitrakoot and Jhansi
- These nodes lie in the primary and secondary catchment area of major National Highways and Expressways

Uttar Pradesh Defence Industrial Corridor

- 57% of Eastern Dedicated Freight Corridor and 15% of Delhi-Mumbai Industrial Corridor lies in Uttar Pradesh
- 47 National Highways spanning 11,737 kms (7.5% of the total NH network) connects Uttar Pradesh to 9 neighbouring states and other parts of the country
- Largest Rail network spanning 8949 kms of rail track
- 5 domestic and 2 international airports connecting 51 cities domestically and 12 internationally. 14 projects under Regional Connectivity Scheme Phase 1, 2 & 3



Defence Sector in Uttar Pradesh

- **Home to prominent DPSUs**
  - HAL, Bharat Electronics Ltd, Troops Comfort Limited, Advanced Weapons and Equipment India Limited, Gliders India Limited

- **Existing R&D and Innovation Base**
  - Renowned academic institutions like IIT-K, IIT-BHU etc.
  - Key R&D institutes like DRDO, ASERDC-HAL
  - Well known Forensics Science Laboratories

- **Strong ancillary base**
  - Home to country's largest MSME base
  - Huge number of repairing/servicing, engineering and spare part manufacturing industry

- **Large skilled manpower base**
  - 56% of UP's 220 million population in the working age group
  - UP has 53 Universities, 4345 colleges and 168 Polytechnics

Summary of UP Defence Industrial Corridor Investments

S.No.	Node	Procured land (in Ha)	Investment Promised (In Crs) from total 57 MoUs
1	Jhansi	1034.6	4,253.3
2	Chitrakoot	101.7	-
3	Kanpur	184.5	477.58
4	Aligarh	83.4	2,037.75
5	Lucknow	141.9	1,094
6	Agra	Phase II	200
	Total	**1546.1**	**8,061.63**

Total Industrial MoUs – **63**

Investment realised from 25 MoUs through land allotments:

Private industries – Rs. 1827.68 Crs.
DPSUs – Rs. 700 Crs (In 5 – 7 Years)
Total – Rs.2527.68 Crs.

Infrastructure Development Timeline

Aligarh – By March 2022
Kanpur – By June 2022
Jhansi, Lucknow & Chitrakoot – Dec 2022
Agra – 2024

Bharat Dynamics Limited

- Land allotted to Bharat Dynamics Limited at Jhansi node
- The Missile Propulsion System manufacturing unit will be set up in 183 hectares
- Initial investment of INR 400 crs in the facility
- Foundation laying done on 19 Nov 2021
- Creation of employment opportunities:
  - 150 direct employment
  - More than 500 indirect employment
- Encouragement and support to large number of vendor MSME units
- Unit will create a boasting presence in Bundelkhand region

Image Source: The Federal News

Brahmos Aerospace

- Land allotted to BrahMos Aerospace Limited at Lucknow node
- The modern, state-of-the-art production facility in an 80-hectare area
- Investment of INR 300 crs in civil infrastructure, mechanical equipment and allied facilities in 5 to 7 years
- Over 200 small, medium and large defence enterprises, institutions and laboratories associated with the missile programme
- Creation of employment opportunities:
  - 500 direct employment for Engineers and Technicians
  - 5,000 indirect employment
  - 10,000 Ancillary Unit employment for skilled, semi-skilled and non-skilled workers
- Boost existing and upcoming MSME base in the Corridor

Image Source: Economic Times

Image Source: European Security & Defence

State Defence & Aerospace Policy

- **25% Land subsidy to all Anchor units Capital subsidy:**
  - For Anchor Units – upto 10% (ceiling 10 Crs) and upto 15% (ceiling INR 15 Crs) in Bundelkhand
  - For MSME– upto 5% (ceiling 5 Crs) and upto 7.5% (ceiling INR 7.5 Crs) in Bundelkhand
- 100% Stamp Duty exemption for Anchor Units
- Provision of transport subsidy for raw materials & finished goods
- Provision of land to support Common Facility Centres
- Dovetailing with the Uttar Pradesh Start-up Policy to promote Start-ups
- Support for Quality certification, trademark registration and Patent registration
- Support for training of personnel

Centrally Sponsored Scheme

- Department of Defence Production has formulated a draft scheme to encourage investments in Defence Industrial Corridors

Salient Features

- Matching share between centre and state in 50:50 ratio
- 15% capital subsidy upto a maximum of Rs. 200 Crs. to first 12 units in the corridor with new or incremental capital investment in excess of Rs. 500 Crs.
- 10% capital subsidy for 30 units with new or incremental capital investment between Rs. 100 Crs. and Rs. 500 Crs.
- Setting up of 11 Technology Hubs(THs) and Common Facility Centre(CFC) in and around existing nodes of Defence Industrial Corridors

Assistance to investors

- As part of Ease of Doing Business, all approvals and clearances being channeled though Nivesh Mitra
- Development of online Building Approval System to fast-track grant of Building Permits
- Hand holding for project planning, licensing, clearances, collaborations and financing options to industrial units especially to the new ventures
- Dissemination of information to investors on various developments in the sector along with access to sector specific Seminars and Webinars
- Partnering with banking institutions such as SBI and SIDBI to assist in obtaining required financial assistance
- SIDBI has created special schemes for MSMEs with longer repayment terms, varied interest rates, lucrative financing options etc
- SBI is also in the process of creating tailor made banking products for MSMEs of the Corridor

Aligarh Node - Brief Snapshot

- Aligarh is the first fully populated node of UP Defence Industrial Corridor with 74 hectares land leased to 19 companies
- Located on Aligarh-Palwal State highway and 36 kms from Yamuna Expressway
- Aligarh has a developed ecosystem of precision manufacturing, diecasting and Research labs
- Node being developed as a base for Unmanned Aerial Systems, Component manufacturing Precision component, Small arms and metal works
- Land procured and allotted – 183 Acres, Additional land under procurement process
- Total MoUs – 29, worth Rs. 2,037 Crs.
- Realised MoUs – 19, worth Rs. 1,245.75 Crs.

Kanpur Node - Brief Snapshot

- Located in Sarh Tehsil and lies on State Highway 17
- Kanpur has an existing robust ecosystem of leather and textile industry
- Home to 3 of the 7 DPSUs created after corporatisation of OFB
- Node being developed as a base for Special clothing, Bullet proof jackets, Ammunition, Components and Parachutes
- Land procured 183 Ha, Allotted – 4 Ha
- Total MoUs – 7,worth Rs. 477.58 Crs.
- Realised MoUs – 2, worth Rs. 58.58 Crs.

Jhansi Node - Brief Snapshot

- Located on the Erach – Benda Road and lies 8 kms from NH27
- The upcoming Bundelkhand Expressway is 40 kms from the node
- Jhansi node is being developed as a base for land systems, missile systems, small arms and ammunition and Research & Development
- CFC of MSME will come up in Jhansi under MSME policy of Central Government
- Land procured 1034 Ha, Allotted – 183 Ha
- Total MoUs – 7, worth Rs. 4200 Crs
- Realised MoUs – 2, worth Rs. 603 Crs

Lucknow Node - Brief Snapshot

- Located in Village Bhatgaon, Sarojini Nagar Tehsil
- Node lies in proximity to the under construction Kisan Path and approximately 7.5 kms from NH 27
- Node being developed as an Aerospace hub
- Land procured and allotted - 141 Ha
- A DRDO research centre is being set up in Lucknow
- Total MoUs - 11, worth Rs. 1,094 Crs
- Realised MoUs – 2, worth Rs. 620 Crs

Existing manufacturing base

Uttar Pradesh has several units in the public and private sector which are involved in manufacturing in the defence and aerospace sector. Public sector units are the mainstay of the strong local market as they procure materials and essential components from local sources. The public sector heads include 9 Indian Ordnance Factories and 3 Hindustan Aeronaucs Limited (HAL) manufacturing units.

List of Ordnane Factories in Uttar Pradesh	
Troop Comforts Limited	Headquarters, GT Road Kanpur - 208 0135
Gliders India Limited	Headquarters, GT Road Kanpur - 208 005
Advanced weapons and Equipment India Limited	Ordnance Factory Kanpur, Kalpi Road Kanpur - 208 009

Image Source: India Today

Research & Development Ecosystem

Uttar Pradesh has various educational and training institutes engaged in research and development. There are 53 universities, 4,345 colleges, 168 polytechnics in the state including many research institutes, centres of excellence and other professional institutes. The state is the stronghold of specialized institutions like IIT Kanpur and IITBHU. Major institutes like the Public Sector Defence Research and Development Organization, QA and R&D Establishment (DMSRDE), HAL, etc., provide excellent research and development ecosystems for the defence and aerospace sector in Uttar Pradesh. Under HAL, Aerospace system and Equipment R&D Centre (ASERDC) located in Lucknow is using stateofthe art technology and is involved in applied research, design and development of main system & equipment for aircrafts and helicopters. Flight data recorders and other Avionics LRUs are developed in the ASERDC of HAL unit located in Korwa. Apart from this, Forensic Science Laboratories are located in Lucknow and Agra in Uttar Pradesh. 7 divisions of these labs are providing modern techniques & equipment to the UP police.

Hindustan Aeronautics Limited

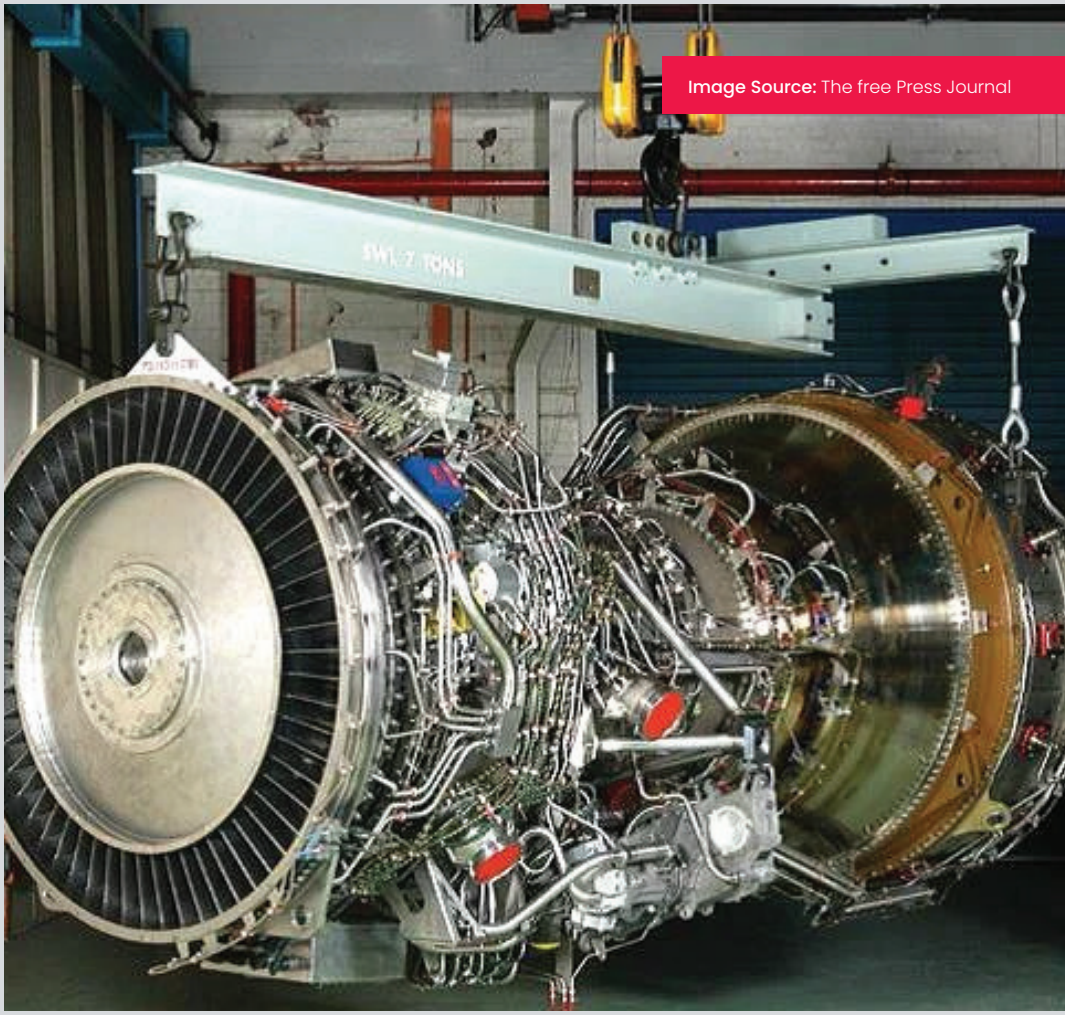


Image Source: The free Press Journal

Achievements

1. India is the world's largest military hardware importer and among the top 5 military spenders. The Indian defence industry is going through a rapid change and indigenous manufacturing is being promoted extensively under the 'Make in India' scheme. The Finance Ministry of India has also announced foreign direct investment (FDI) up to 74 per cent under the automatic route in the defence sector. **The Union Budget 2021-22 allocated USD 49.55 Bn for defence services and other organizations/departments under the Ministry of defence** (excluding defence pensions). The government is encouraging collaborations like joint ventures, partnerships, offsets, etc. with an aim to decrease foreign dependency.

2. To expand the defence manufacturing capability within the country and reducing foreign dependency, **two Defence Industrial Corridors (DICs) in Uttar Pradesh and Tamil Nadu are progressing at a rapid pace.** Uttar Pradesh, Defence Industrial Corridor is a Greenfield project.

3. The state has declared 6 districts (Aligarh, Agra, Jhansi, Lucknow, Kanpur and Chitrakoot) as the defence node. **In order to initiate growth of the manufacturing industry, the state has come out with various incentives and subsidies in its Defence & Aerospace Unit and Employment Promotion Policy.** A large number of industries including MSMEs and start-ups from the non-defence sector in the region would be able to take advantage of the Aerospace & Defence Policy and integrate themselves into the defence manufacturing ecosystem. The state also provides all desired approvals to the Aerospace and Defence manufacturing units through the single window system.

4. Over 1546.35 hectares of land at – Aligarh, Kanpur, Jhansi and Chitrakoot has been procured till date. 25 companies have been alloted land across 4 nodes of the UPDIC.

5. The EDFC and the WDFC have provided the state with huge advantage in logistics and warehousing sector. Strong air connectivity; being augmented further by commercially activating international and domestic airports. The infrastructure being developed at every node is going to be world class and one of its kind in the country.

6. **The state has formulated various polices aimed at supporting the industry:**

- Uttar Pradesh Defence & Aerospace Unit and Employment Promotion Policy, 2019 and Guidelines, 2020
- UP Start-up Policy, 2020
- Industrial Investment and Employment Promotion Policy, 2017
- U.P. Micro, Small & Medium Enterprises Promotion Policy, 2017
- Uttar Pradesh Electronics Manufacturing Policy, 2020
- Post Covid-19 Accelerated Investment Promotion Policy

7. **The UPDIC micro site: http://upeida.up.gov.in/updic/ was launched during Aero India 2021 to provide seamless access and information to all interested investors.** This site provides detailed information about each node including existing industrial base as well as relevant policies of MoD and the state and indigenisation opportunities under various DPSUs and Armed Forces Indigenisation Cells.

8. **Outreach programs through mailers, articles, webinars and seminars have been a significant driving force to spread the word for the corridor.** Some significant webinars were organized with – SIDM, CII, UKIBC, USIBC, ASSOCHAM, FICCI to ensure better propagation of information related to UPDIC among industries.

9. **UPEIDA has been active in international participations to promote UPDIC**, for e.g. the Indo-Russian Defence Industry Corporate Conference in November 2019 in Moscow, Defence and Security Equipment International Seminar in London (DSEIUK) in September 2019. A similar delegation/representation happened at South Korea. These interactions and visits have been a key contributor in creating possibilities and opportunities in terms of Transfer of Technology and bringing FDI into UPDIC.

10. **DEF EXPO 2020 and Aero India 2021 have given Uttar Pradesh a head start. A total of 68, including 11 institutional MoUs have been signed till date, estimating investments worth more than INR 8000 crore.** Out of these, 19 MoUs have been realized bringing investment worth more than INR 1000 crore. On 5 August 2021, UPEIDA signed an MoU with Bharat Dynamics Ltd. for setting up the defence corridor's first anchor unit (investments worth INR 400 crore in phase 1) in Jhansi on 183 hectare of land for the manufacturing of propulsion systems. UPEIDA also signed an institutional MoU with SIDM for increasing the outreach in the industry.

11. UPDIC has joined hands with various central government agencies, training institutes, skill development centers, IITs, academia, financial institutes and armed forces of the country so as to ensure that an excellent ecosystem is available to the new industries being set up in the UPDIC. **UPEIDA has also created two Centres of Excellence (CoE) along with IITK for drones and cloud seeding.**

12. Hon’ble Chief Minister of Uttar Pradesh personally monitors and guides UPDIC project as this is going to be a flagship project for the state of UP in the field of Aerospace and Defence.

13. Uttar Pradesh may soon be the state where the next generation BRAHMOS (BRAHMOS-NG) and other missiles will be manufactured as the Uttar Pradesh Defence Industrial Corridor (UPDIC) has started taking shape at its six nodes. **BRAHMOS Aerospace, the joint venture of Defence Research and Development Organisation (DRDO) of India and NPO Mashinostroeyenia (NPOM) of Russia, has proposed to set up the modern production facility in Lucknow with an investment of INR 300 crore.**

14. On September 14, 2021, Hon'ble Prime Minister Shri Narendra Modi visited the Aligarh node of Defense Corridor and reviewed the progress.
On November 12, 2021, Hon'ble Defence Minister and Hon'cle Chief Minister of Uttar pradesh held a massive consultative session with industry giants, newly formed DPSUs and 100+ MSME and startups in the Defence and Aerospace sector. On November 19, 2021, Hon'ble Prime Minister laid down the foundation stone of Bharat Dynamics Ltd's project in Jhansi node of UPDIC.

15. **Total MoUs**

(a) Industry MoUs - 63 Nos (b) Institutional MoUs - 11 Nos **Total** - 74 MoUs

16. **Land Allocation to 25 companies - 357.4 Hectares**

S No.	Company's Name	Land Allotted (In Ha)	Proposed Investment Intent (in Crs.)	Domain
ALIGARH NODE				
1	Allen & Alvan Pvt Ltd	8	30.75	Drones
2	Nitya Creations India	1.5	12	Precision components
3	PBM Insulations Pvt. Ltd	0.4	4	Insulated mats
4	M/s Jai Sai Anu Overseas Pvt. Ltd	4.5	100	Precision components
5	Deep Explo Equipments Pvt Ltd	1	10.35	Metallic parts for grenades and explosives
6	Kobra Industries	0.25	1	Defence packaging
7	Shridha Udyog	1	2.7	Parts for aerospace sector
8	M/S Precision Products	1	2.7	Precision components

S No.	Products	Land Allotted (In Ha)	Proposed Investment Intent (in Crs.)	components
ALIGARH NODE				
9	Werywin Defence Pvt. Ltd	1.671	65	Small arms
10	Advanced Fire & Safety	1	3	Fire safety equipment
11	Navraj Metal Works	1.6	20	Tungsten rods
12	Ancor Research Labs LLP	10	550	Drones
13	P2 Logitech Pvt Ltd	2	90	Optical sights
14	New Space Research & Technologies Pvt Ltd	3.5	35	UAVs
15	Crimson Energy Experts Pvt. Ltd.	0.58	11	Navigational devices
16	Tractrix Opto Dynamic LLP	1.4	40	Precision and optical instruments
17	Syndicate Innovations International Ltd	10	150	Small arms and ammunition

S No.	Products	Land Allotted (In Ha)	Proposed Investment Intent (in Crs.)	components
ALIGARH NODE				
18	Royal Sales Pvt. Ltd.	2	20	Precision components
19	Milkor Defence Pvt. Ltd.	4	98.25	Defence Weapon Manufacturing
KANPUR NODE				
20	Kungarmour Pvt Ltd	1	20	Defence clothing
21	Adhunik Material and Science Pvt. Ltd.	3	38.5	Special material for bulletproof jackets
JHANSI NODE				
22	Delta Combat Systems	15	203.3	Small arms
23	Bharat Dynamics Ltd	183	400	Manufacturing of Propulsion System

S No.	Products	Land Allotted (In Ha)	Proposed Investment Intent (in Crs.)	components
LUCKNOW NODE				
24	DRDO Brahmos Aerospace	80	300	Manufacturing of Brahmos Missile
25	Aerolloy Technologies Ltd	20	320.25	Titanium & super alloys components, Metal, sub-systems
	TOTAL	**357.401**	**2527.68**	

Opportunities available

With excellent infrastructure, facilities and a favourable ecosystem, this policy is intended to attract investment in the following areas of defence and aerospace:

1. Defence Technical Park in the defence corridor nodes like Kanpur, Jhansi, Aligarh, Chitrakoot, Agra and Lucknow and its expansion in other districts.
2. Expansion and participation of defence public sector units.
3. Establishment of Aerospace Technology Park with the possibility of expanding it in defence corridor nodes like Kanpur, Jhansi, Aligarh, Chitrakoot, Agra and Lucknow and other districts.
4. Setting up of Testing and Validation Centres, including a firing range for Artillery and other military weapons/ equipment.
5. Development of UAV/drone format manufacturing and testing facilities.
6. Manufacturing or assembling of aeroplanes, helicoptersand maintenance facilities.
7. Combination of components of defence/military/aerospace mechanical vehicles, their components and establishment of ancillary units.
8. Weapons and sensor equipment used in police modernization and small intensity conflicts.
9. Services related to electronics and information technology in Agra, Gautam Budh Nagar, Ghaziabad, etc.
10. Metal precision work at engineering centre-Aligarh and foundry work in Agra.
11. Manufacturing centres for leather, textiles, shoes and other technical ancillary equipment for the defence and aerospace sector.
12. Centre for manufacturing of arms, weapons systems, ammunition explosives and ancillary components.
13. Establishment of special food manufacturing and packaging centres related to defence and aerospace.

Def corridor, logistic hub to boost UP's growth: Mahana

'Govt Policies Attracted Big Investments'

TIMES NEWS NETWORK

Lucknow: Industrial development minister Satish Mahana on Wednesday listed out the government's achievements, including the defence corridor, airports and expressways, foreign investments and sectoral policies.

lize the vast SME ecosystem in the state. Around 2.5 lakh employment and a total investment of Rs 50,000 crore are expected through it. It has been planned over 5,000 hectare of which 1,445 hectare has been acquired. All available land in the Aligarh node has been allotted to 19 companies, which are expected to invest around Rs 1,245 crore," Mahana said.

The state government held an investors' summit in 2018 and 43% of the leads generated then, worth nearly Rs 1.84

He added that 218 projects worth Rs 51,923 crore have started commercial production while 130 projects worth Rs 37,376 crore are under implementation.

The state government, he said, had not just created 20 sectoral policies to promote investment under sectors such as energy, logistics and food processing, it had also acquired large tracts of land under a land bank which is being made available to investors without wasting time.

enabled by the single window portal Nivesh Mitra which is offering more than 340 services of 27 departments to investors," Mahana said.

A leather park is coming up in Unnao, and a logistic hub and heritage city is being set up next to the Yamuna Expressway, he said.

The Hiranandani Group is setting up data centre in Greater Noida for Rs 6,000 crores, MAQ India Pvt Ltd is setting up a software development unit in Noida worth Rs 252 crore and Dreamtech Electronics India Pvt Ltd is setting up an

Way Forward

The upcoming defence industrial corridor in Uttar Pradesh is a new step towards achieving self-reliance. It will bring with it a multitude of benefits not just for the state but for India as well.

Apart from reviving the PSU base, itshall mobilize the vast SME ecosystem in the state. The proposeddefence corridor will play a key role in the state's development by bringing in jobs and investments.Bundelkhand will benefit directly from the six-lane highway which the Uttar Pradesh Government has proposed to build up from Chitrakoot to Jhansi and will provide necessary infrastructural support for the Defence Production Units. Traditionally, UP has possessed skilled workmanship in metals, besides having a vast defence manufacturing base in Kanpur with almost 6 public sector undertakings manufacturing arms and defence related wares. However, these were never leveraged on big scale by successive governments, but this Defence Corridor will definitely provide the needed impetus to the Defence Manufacturing in Uttar Pradesh & will play a significant role in industrial development of Uttar Pradesh simultaneously providing job opportunities to the potential unemployed youth workforce. The Center has already announced to set up common facility centers in the corridor to incentivise private companies set up shop and cut their production cost. Moreover, availability of land at comparatively reasonable rates and adequate human resourceswill help the companies interested in setting up their units here.

Thus, Uttar Pradesh Defence Industrial Corridor will play an important role in changing India's image as the largest importer of defence equipment to a major exporter of defence-related hardware.Uttar Pradesh is getting all set to become self-reliant in the manufacturing of defence equipment in the future.The goals and vision of the state are also very clear and the progress it has made till date is commendable. This project will definitely serve as an engine for economic development and growth of the regions of Uttar Pradesh.

Uttar Pradesh Defence Industrial Corridor:

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